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Student Number

2015
HSC course
Mathematics
EXTENSION 1
25th May 2015

General Instructions

- Reading time – 5 minutes
- Working time - 1 hour
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Start a new booklet for each question

Total Marks – 45

Question	Marks
1	/14
2	/15
3	/16

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 20% of the Higher School Certificate Course Assessment

	Questions	Marks
1.	a) If $f(x) = 2 \sin^{-1} 3x$ then find: i) the domain and range of $f(x)$. ii) $f\left(\frac{1}{6}\right)$. iii) $f'\left(\frac{1}{6}\right)$. b) For what value of x will the geometric series have an infinite sum? $\left(\frac{x}{2x-1}\right) + \left(\frac{x}{2x-1}\right)^2 + \left(\frac{x}{2x-1}\right)^3 + \dots$ c) How many straight line arrangements can be made from the word BALANCES if: i) there are no restrictions? ii) the letters B and S are at either end? iii) the 3 vowels are next to each other? d) Prove by Mathematical Induction that $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n \text{ for all integers } n \geq 1.$	2 1 2 2 1 2 1 3

	Questions	Marks
2	a) i) Sketch the function $y = f(x)$ where $f(x) = (x-1)^2 - 4$ clearly showing all intercepts on the coordinate axes (use the same scale on both axes). ii) What is the largest positive domain of $f(x)$ for which $f(x)$ has an inverse $f^{-1}(x)$? iii) Sketch the graph of $y = f^{-1}(x)$ on the same number plane as in part i). b) i) Show that $\cos^{-1} \frac{3}{5} - \tan^{-1} \left(\frac{-3}{4} \right) = \frac{\pi}{2}$ ii) Differentiate $x \tan^{-1} x$. iii) Hence evaluate $\int_0^1 \tan^{-1} x \, dx$ c) There are 12 DVD's arranged in a row on a shelf in Mrs Jackson's study. There are 5 copies of Superman 1, 4 copies of Avengers and 3 copies of Gladiator. i) How many arrangements of the DVD's are possible? ii) How many different arrangements are there if the DVD's with the same title are grouped together? d) Consider the word EXERCISES. What is the probability of an arrangement beginning with the letters CR or RC.	2 1 1 3 1 3 1 1 2

	Questions	Marks
3	a) Kyle borrowed \$26 000 ($P$) at an interest rate of 9% p.a. compounded monthly. She is to repay the loan in 8 equal instalments over 2 years. That is, she will make repayments of \$$R$ at 3 monthly intervals. Let $K = (1.0075)^3$ and A_n = the amount owing on the loan immediately after the nth repayment. i) Show $A_n = PK^n - \frac{R(K^n - 1)}{K - 1}$. ii) Show that the value of the repayments \$$R$ is \$3590.20. iii) When Kyle made her 7th repayment she repaid an additional \$2000. That is, she repaid \$5590.20. Determine the amount by which her final repayment will be reduced as a result of the additional \$2000 in her 7th repayment. b) The region in the first quadrant bounded by the curve $y = 2\tan^{-1} x$ and the y-axis between $y = 0$ and $y = \frac{\pi}{2}$ is rotated about the y-axis. Find the exact volume of the solid of revolution formed. c) Prove by Mathematical Induction that $3^n + 4^n < 5^n$ for all positive integers $n \geq 3$. d) Find the number of ways of seating 5 boys and 5 girls around a round table if: i) There is no restriction where each person sits. ii) A particular girl (L) wishes to sit between 2 particular boys B and G. iii) Two particular people (G and J) do not wish to sit together. END OF ASSESSMENT	2 2 2 3 3 1 1 2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Question 1

a) i) Domain: $-1 \leq 3x \leq 1$ Range: $-\pi \leq y \leq \pi$
 $-\frac{1}{3} \leq x \leq \frac{1}{3}$

ii) $f(\frac{1}{6}) = 2 \sin^{-1} \frac{1}{2}$
 $= 2 \times \frac{\pi}{6}$
 $= \frac{\pi}{3}$

iii) $f(x) = 2 \sin^{-1} 3x$
 $f'(x) = 2 \times \frac{3}{\sqrt{1-9x^2}}$
 $f'(\frac{1}{6}) = \frac{6}{\sqrt{1-9 \times \frac{1}{36}}}$
 $= 4\sqrt{3}$

i) Domain
 1 Range

ii) 1 for correct answer

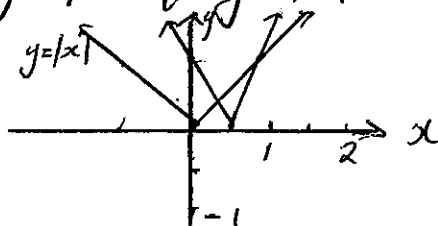
iii) 1 for correct differentiation
 1 for correct substitution

b) $r = \frac{x}{2x-1}$, For sum to infinity
 $|r| < 1$

$\left| \frac{x}{2x-1} \right| < 1$

$|x| < |2x-1|$

Consider graph of $y=|x|$ and $y=|2x-1|$



$y=x$ and $y=-2x+1$ intersect at $x=\frac{1}{3}$

$y=x$ and $y=2x-1$ intersect at $x=1$

$\therefore |x| < |2x-1|$ occurs where
 $x < \frac{1}{3}$ and $x > 1$

$\therefore$ an infinite sum exists where $x < \frac{1}{3}$ and $x > 1$

1 for showing graphs of
 $y=|x|$ and
 $y=|2x-1|$

or
 a method that works from
 $|x| < |2x-1|$
 (don't give mark for this line only)

1 for answer

c) i No Restrictions = $\frac{8!}{2!}$

ii Arrangements = $\frac{2! \times 6!}{2!} = 6!$

iii When 3 vowels are together = $\frac{6!}{3}$

1 for $\frac{8!}{2!}$

1 for $2! \times 6!$

2 for working to final answer

1 for answer

Question 1

d) For $n=1$ LHS = 1, RHS = 1
 $\therefore$ true for $n=1$

Assume true for $n=k$

ie $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$ (1)

Required to show for $n=k+1$

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1)2^k = 1 + k \times 2^{k+1}$$

$$\begin{aligned}\text{Now, LHS} &= 1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1)2^k \\ &= 1 + (k-1)2^k + (k+1)2^k \quad (\text{sub from (1)}) \\ &= 1 + 2^k [(k-1) + (k+1)] \\ &= 1 + 2^k \times 2k \\ &= 1 + k \times 2^{k+1} \\ &= \text{RHS}\end{aligned}$$

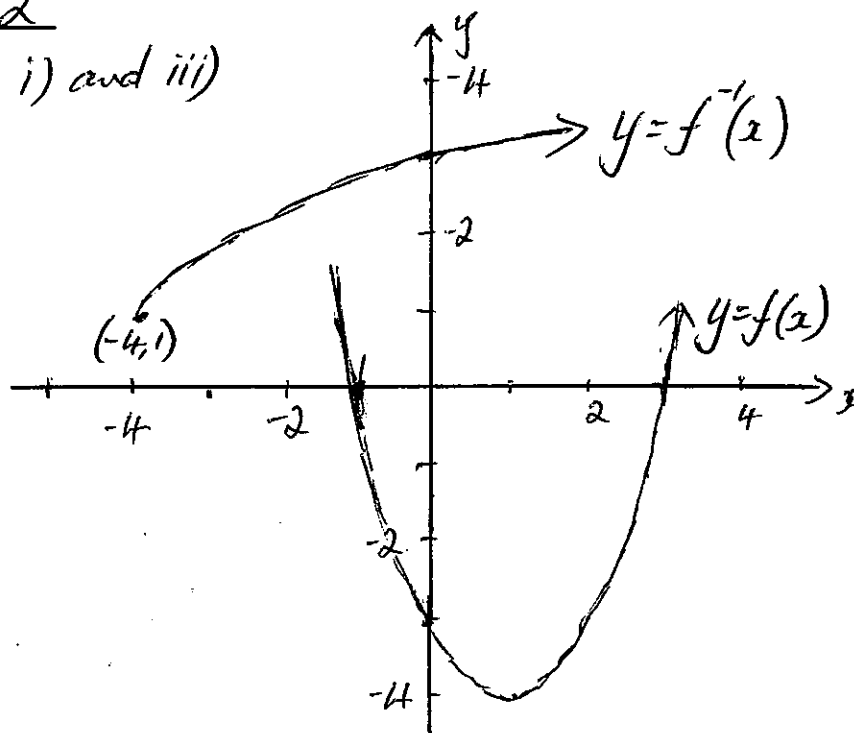
If the result is true for $n=1$ and proven true for $n=k+1$ after assumed to be true for $n=k$ then it is true for all positive integers of n .

1 for showing
true for $n=1$
1 for writing
the correct
assumption

1 for working
through the
necessary steps
to show true
for $n=k+1$

Question 2

a) parts i) and iii)



i) 1 mark for shape
1 mark for intercepts,
and showing $(1, -4)$

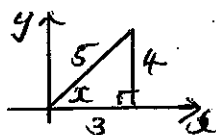
iii) 1 mark for showing $(-4, 1)$
and passing through $(0, 3)$

ii) 1 for answer

ii) $x > 1$

b) i) $\cos^{-1}\frac{3}{5} - \tan^{-1}\left(-\frac{3}{4}\right) = \frac{\pi}{2}$

let $x = \cos^{-1}\frac{3}{5}$
 $\therefore \cos x = \frac{3}{5}$



$\therefore x - y = \frac{\pi}{2}$

Take sine of both sides

i.e. $\sin(x - y) = \sin\frac{\pi}{2}$

LHS = $\sin(x - y)$

= $\sin x \cos y - \cos x \sin y$

= $\frac{4}{5} \times \frac{4}{5} - \frac{3}{5} \times \left(-\frac{3}{4}\right)$

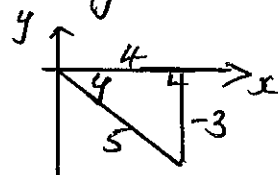
= 1

RHS = $\sin\frac{\pi}{2} = 1$

Thus LHS = RHS

$\therefore \cos^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(-\frac{3}{4}\right) = \frac{\pi}{2}$

let $y = \tan^{-1}\left(-\frac{3}{4}\right)$
 $\therefore \tan y = -\frac{3}{4}$



1 for using $\cos$ and $\tan$ to write
 $\cos x = \frac{3}{5}$ and
 $\tan y = -\frac{3}{4}$

1 for diagrams
and writing
 $\sin(x - y) = \sin\frac{\pi}{2}$

1 for working
through this
to show

$\cos^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(-\frac{3}{4}\right) = \frac{\pi}{2}$

Question 2

c) i) Number of arrangements = $\frac{12!}{3!4!5!}$

1 for answer

ii) Number of arrangements = $3!$

1 for answer

d) Total Number of arrangements = $\frac{9!}{3!2!}$

Number of arrangements
starting with CR or RC = $\frac{2!8!}{3!2!}$
 $= \frac{8!}{3!}$

1 for $\frac{8!}{3!}$

$P(\text{start with CR or RC}) = \frac{8!}{3!} \div \frac{9!}{3!2!}$
 $= \frac{8! \times 2}{9!}$

1 for final
answer

d) ii) $y = x \tan^{-1} x$
 $\frac{dy}{dx} = x \left(\frac{1}{1+x^2} \right) + \tan^{-1} x$
 $\frac{d}{dx}(x \tan^{-1} x) = \frac{x}{1+x^2} + \tan^{-1} x$

1 for the
answer

iii) $\int_0^1 \tan^{-1} x \, dx = \int_0^1 \frac{d}{dx}(x \tan^{-1} x) \, dx - \int_0^1 \frac{x}{1+x^2} \, dx$

1 for writing
the definite
integrals

$$= [x \tan^{-1} x]_0^1 - \left[\frac{1}{2} \ln(1+x^2) \right]_0^1$$

1 for substituted

$$= \tan^{-1} 1 - 0 - \left[\frac{1}{2} \ln 2 - \frac{1}{2} \ln 1 \right]$$

1 for answer

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Question 3

a) i) $A_1 = P(1.0075)^3 - R$

$$\begin{aligned} A_2 &= \{P(1.0075)^3 - R\}(1.0075)^3 - R \\ &= P(1.0075)^6 - R[(1.0075)^3 + 1] \\ &= PK^2 - R(K+1) \end{aligned}$$

$$\begin{aligned} A_n &= PK^n - R(K^{n-1} + K^{n-2} + \dots + 1) \\ &= PK^n - R \times \frac{a(r^n - 1)}{r - 1} \\ &= PK^n - R \times \frac{1(K^n - 1)}{K - 1} \\ &= PK^n - \frac{R(K^n - 1)}{K - 1} \end{aligned}$$

ii) When $n=8$, $A_n=0$

$$\therefore PK^8 - \frac{R(K^8 - 1)}{K - 1} = 0$$

$$\therefore 26,000 \times (1.0075)^{24} = \frac{R(1.0075^{24} - 1)}{1.0075^{23} - 1}$$

$$R = \$3590.20$$

$$\begin{aligned} \text{iii) } A_7 &= 26000(1.0075)^{21} - \frac{3590.20(1.0075^{21} - 1)}{1.0075^3 - 1} \\ &= \$3510.64 \end{aligned}$$

Because Kyle repaid an additional \$2000 for the 7th repayment then the amount owing at the start of the last 3 months is

$$3510.64 - 2000 = 1510.64$$

$$\begin{aligned} \text{Last payment} &= 1510.64 \times 1.0075^3 \\ &= \$1544.88 \end{aligned}$$

$$\begin{aligned} \text{Last payment is reduced by} &= 3590.20 - 1544.88 \\ &= \$2045.32 \end{aligned}$$

1 for working towards A_2

1 for writing A_n from A_2 and working through to the final result

1 for correct substitution

1 for working towards correct answer

1 for obtaining \$1510.64

1 for working towards \$2045.32

Question 3

$$b) y = 2 \tan^{-1} x$$

$$\frac{1}{2} y = \tan^{-1} x$$

$$\tan\left(\frac{1}{2} y\right) = x$$

$$V = \pi \int_0^{\frac{\pi}{2}} \tan^2\left(\frac{1}{2} y\right) dy$$

$$= \pi \int_0^{\frac{\pi}{2}} \sec^2\left(\frac{1}{2} y\right) - 1 dy$$

$$= \pi \left[2 \tan\left(\frac{1}{2} y\right) - y \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\left(2 \tan \frac{\pi}{4} - \tan 0 \right) - \left(\frac{\pi}{2} - 0 \right) \right]$$

$$\therefore \text{Volume} = \frac{1}{2} \pi (4 - \pi) \text{ units}^3.$$

1 for writing
 $x = \tan\left(\frac{1}{2} y\right)$

1 for definite
integral

1 for working
towards
final answer

c) For $n=3$

$$\text{LHS} = 3^3 + 4^3$$

$$= 91$$

$$\text{RHS} = 5^3$$

$$= 125$$

LHS < RHS $\therefore$ true for $n=3$

Assume true for $n=k$

$$\text{ie } 3^k + 4^k < 5^k$$

$$\text{ie } 5^k > 3^k + 4^k \text{ — (1)}$$

Required to show true for $n=k+1$

$$\text{ie } 5^{k+1} > 3^{k+1} + 4^{k+1}$$

$$\text{Now, LHS} = 5^{k+1}$$

$$= 5 \times 5^k$$

$$> 5 \times (3^k + 4^k) \text{ substitution from (1)}$$

$$= 5 \times 3^k + 5 \times 4^k$$

$$> 3 \times 3^k + 4 \times 4^k$$

$$= 3^{k+1} + 4^{k+1}$$

$$= \text{RHS}$$

$$\text{ie } \text{LHS} > \text{RHS}$$

~~The~~ the result is true for $n=3$ and
proven true for $n=k+1$ after assuming
true for $n=k$. $\therefore$ it is true for all
positive integer $n \geq 3$.

1 for showing
true for $n=3$

1 for assumption

1 for showing
true for
 $n=k+1$.

Working must
be shown.

Question 3

d) i) No restriction = $9!$

1 for answer

ii) Number of ways
girl is between = $2 \times 7!$
2 boys

1 for answer

iii) 2 particular not
wishing to be together = $9! - \text{them together}$
= $9! - 2 \times 8!$
= $282,240$

1 for subtract
 $2 \times 8!$
1 for answer